

The Seed and the Soil: A Longitudinal Network and Resource-Based View Analysis of Entrepreneurial Ecosystems in MENA

Abstract

Entrepreneurial Ecosystems (EEs) are vital for fostering innovation and economic growth in emerging markets. Their organization is essential. To run for the first time an internal analysis on EE organization, we frame an innovative research design leveraging network analysis associated with the Resource-Based View (RBV) and VRIO frameworks to observe EE resource complementarities. We compare six MENA countries—Egypt, Morocco, Tunisia, Qatar, Saudi Arabia, UAE—focusing on their structure and evolving characteristics. We webscraped 617,058 observations from 2013 to 2024 on 1,603 EE actors. The research shows that both the UAE and Qatar have strong, interconnected networks, making it easier to access resources, presenting an abundant soil for entrepreneurs. In contrast, Egypt and Saudi Arabia exhibit fragmentation and poor resource complementarity; Morocco and Tunisia display volatility in cohesiveness and efficiency. These findings highlight how cooperation occurs among EE organizations for effective resource distribution. The study helps explain how EE acts as a value chain for entrepreneurs. It also provides useful advice for policymakers and practitioners who want to encourage innovation and address social and economic issues in developing countries.

Keywords: Middle East and North of Africa (MENA); Entrepreneurial Ecosystem; Structural Analysis; Resource-Based View (RBV)

Introduction

To promote entrepreneurship, innovation, employment, and economic growth, EE is a crucial framework for research and management (Meshram and Rawani, 2019; Spigel and Harrison, 2018). EEs are complex networks of interconnected and embedded actors who collaborate to provide both financial and non-financial support to start-up companies (Korsgaard et al., 2022). These actors

include institutions, accelerators, incubators, venture capital, and practical services (Malecki, 2018). We characterize EEs as territorial or regional phenomena, focusing networks and resources on local entrepreneurial dynamics. Leveraging EE structural efficiency measures in emerging territories has the potential to inform on how to solve socioeconomic issues, such as unemployment and inequality, since competitiveness and economic growth issues have globally dramatically grown (Mago and van der Merwe, 2023). Effective production and distribution by those actors of suitable entrepreneurial financial and non-financial resources are required to support entrepreneurial dynamics. The purpose of this study is to investigate techniques for examining complementary resources and identifying a structural pattern that permits appropriate theorization.

Entrepreneurs gain access to essential resources such as finance, talent, and knowledge through the supportive environments provided by ecosystem elements (Stam, 2015). Recent research highlights the importance of ecosystem elements in shaping entrepreneurial dynamics, shifting the focus towards the qualitative development of new ventures instead of merely increasing their quantity (Szerb et al., 2019). Studies indicate that the interplay between economic growth and entrepreneurial activity is significantly affected by robust EEs (Content et al., 2020). Currently, challenges persist in the development and sustainability of these EEs (Mago and van der Merwe, 2023).

From an economic perspective, EEs encompass several elements that influence growth and entrepreneurial endeavors. Demand, talent, finance, physical infrastructure, and formal institutions are essential components (Leendertse et al., 2021; Queissner et al., 2024). Their association underscores the systemic nature of EEs. Survival rates for ventures are often elevated among high-quality EEs characterized by a supportive culture, financial resources, human capital, innovative capacity, and institutional support frameworks (Vedula and Kim, 2019). Still, the way EE is viewed under economic lenses doesn't take into account the different ways EE actors structurally collaborate, especially in developing countries, which leads to zooming in on the way EEs are organized. From an organizational perspective, variations in entrepreneurial ecosystem structure influence the relationship between economic growth and entrepreneurship (Content et al., 2020). But research does not actually

propose methods to measure EE organization. Measuring EE through innovative perspectives is crucial for scientific progress and policy formulation, underscoring the necessity for the development of comprehensive metrics and indices. Understanding these ecosystemic structural processes can guide targeted measures to support entrepreneurial economies.

In this study, we examine the EEs of six MENA nations: Egypt, Morocco, Tunisia, Qatar, Saudi Arabia (KSA), and the United Arab Emirates (UAE) (Van Horne et al., 2013). Because of their distinct political, economic, and historical backgrounds, each of these nations has distinctive traits (Zengyu Huang et al., 2013). We seek to identify the structural and dynamic features of these ecosystems (Mack and Mayer, 2016) as well as patterns by utilizing a network analysis approach and the RBV metrics, a first in our best knowledge. This comparative analysis sheds light on how these ecosystems have evolved over time and what makes them competitive, with an emphasis on how resource dynamics and network interactions affect entrepreneurial success.

Networks play a central role in facilitating the flow of information, resources, and knowledge among the various actors within an ecosystem (Komlósi et al., 2022; Scott et al., 2022). Entrepreneurs, for instance, depend heavily on their networks for accessing capital, market opportunities, and expertise. Investors, mentors, and other ecosystem actors also rely on networks to identify promising opportunities, assess risks, and build synergies (Sullivan and Ford, 2014). The structure of these networks—comprising both formal and informal relationships—determines how effectively the ecosystem functions. A strong network, with deep and widespread connections, allows for the rapid exchange of information, fostering innovation and entrepreneurship. In contrast, fragmented or weak networks may hinder the effective flow of resources, limiting entrepreneurial dynamics (Guéneau et al., 2023; Huggins and Thompson, 2015).

By using network analysis on the EEs of the six MENA countries, we get insights into the flow of resources, the centrality of key actors, and the strength of collaborative links, and more generally, we add to our knowledge and fill the gap of research in this region. In addition, network analysis helps uncover the potential for knowledge spillovers (Balcilar et al., 2023), where actors can benefit from

sharing information and expertise. Ecosystems with dense, well-connected networks are often more resilient to economic crises, as entrepreneurs can turn to their networks for alternative sources of support (Zengyu Huang et al., 2013). On the other hand, ecosystems with fragmented or weak networks may struggle to recover from disruptions, as actors are unable to share resources effectively (Thompson-Whiteside et al., 2023).

The RBV and Dynamic Capabilities (DC) are complementary perspectives that explain an organization's competitiveness and, by extension, the ecosystem as a form of organization (Granovetter, 1985; Powell et al., 1996; Wigren-Kristoferson et al., 2022). RBV focuses on valuable, rare, inimitable, and organized resources (VRIO), including knowledge, human, physical, technological, and organizational resources (Newbert, 2014). DCs emphasize a firm's ability to reconfigure resources in changing environments, encompassing marketing, operational, innovative, and alliance capabilities. Network theory bridged with RBV and DC allows examining resource distribution through network configuration (Burt and Soda, 2021). While competitive dynamics studies external moves, RBV analyzes internal capabilities (Chen et al., 2021). For digital firms, sustainable competitive advantage relies on dynamic capabilities to innovate continually (Cuthbertson and Furseth, 2022). Combining such perspectives could lead to a more thorough knowledge of EE behavior and modularity as well as the underlying causes of its spillovers.

The RBV measurements offer a more thorough comprehension of the generation, distribution, and accessibility of resources in EEs (Godley et al., 2021). Redundancy ratio, accessibility index, alignment ratio, and average pairwise complementarity are among those measurements. The redundancy ratio gauges the degree of resource duplication throughout the ecosystem. While a low redundancy ratio denotes a shortage of vital resources, a high ratio points to inefficiencies in resource distribution. The accessibility index gauges how simple it is for actors to obtain important resources, such as capital or talent, which is essential for business owners trying to grow their companies. The alignment ratio evaluates the degree to which ecosystem participants share strategic objectives and values, which might impact collaboration and synergy. Last but not least, the average pairwise

complementarity metric assesses how well various players' resources complement one another, encouraging cooperation and synergies.

Putting network theory and RBV together gives us a complete way to look at the structural and dynamic abilities of EEs. Through studying how resources move and how networks are set up in an ecosystem, this method makes it easier to find important factors that affect an entrepreneur's success or failure. By analyzing the redundancy ratio, accessibility index, and alignment ratio, this method shows how ecosystems adapt to changes in the market, regulations, and policies.

This research investigates the impact of service complementarities on entrepreneurial results and the temporal evolution of each ecosystem using a longitudinal approach (Audretsch et al., 2021). The findings provide valuable insights for governments, investors, and entrepreneurs aiming to enhance the efficacy and sustainability of EEs in the MENA region. This work enhances our understanding of the evolution and operation of EE, providing a foundation for formulating theories and policies to foster entrepreneurial dynamics.

Literature Review

Network theory offers a foundational paradigm for analyzing EEs, highlighting the significance of inter-organizational linkages and network topologies in the formation and functioning of ecosystems (Scott et al., 2022). These networks are essential for resource distribution, information dissemination, and opportunity generation (Alvedalen and Boschmaa, 2017). Studies indicate that ecosystems with varied, heterophilous actors correlate with increased start-up rates, underscoring the importance of network openness (Prokop and Thompson, 2022). The evolution of EEs encompasses cyclical processes of relationship development, characterized by discrete periods of commencement, development, and maintenance (Scott et al., 2022). The durability of connections and the interconnectivity among supporting entities at various phases are essential determinants of ecosystem efficacy (Guéneau et al., 2023). Nonetheless, obstacles persist in formulating explicit analytical

frameworks, incorporating ideas from network theory, and tackling the multi-scalar and dynamic characteristics of EEs (Alvedalen and Boschmaa, 2017)

Research on EEs underscores the essential function of networks in promoting innovation and resource allocation. In the MENA area, networks assist businesses in addressing resource limitations and maneuvering through regulatory frameworks, with professional and international networks proving especially advantageous for innovation (Zali et al., 2013). The configuration and caliber of these networks substantially influence information accessibility and entrepreneurial results (Zengyu Huang et al., 2013). Research indicates that distinct ecosystem configurations are essential for optimal informal, formal, or external networks, highlighting the need for both requisite and adequate components for efficient networking (Komlósi et al., 2022). The dissemination of knowledge within these ecosystems is crucial for evaluating new venture prospects and growth, with developing viewpoints addressing resource constraints and structural deficiencies (Andrade et al., 2022). These studies highlight the significance of network patterns in EEs across several nations and settings.

The RBV is an interesting paradigm for comprehending EEs, emphasizing the resources and skills that foster competitive advantage (Zahra, 2021). The RBV is a strategic perspective emphasizing the role of resources and capabilities in creating a competitive advantage. It views firms as bundles of heterogeneous, immobile resources that can generate economic rents when valuable, rare, inimitable, and non-substitutable (Miller, 2019). The RBV has evolved to include related concepts such as core competencies, strategic assets, and distinctive capabilities (McGee, 2015). Azzone et al. (1995) propose a reference framework for creating a resource measurement system that links resource measures to overall organization performance in terms of value, contributing to the operationalization of RBV concepts. Although the RBV has been extensively utilized in entrepreneurship research, significant disparities exist between the conceptualizations of resources by researchers and entrepreneurs, indicating a necessity for contextual adaptation in the field of entrepreneurship (Kellermanns et al., 2016). Roundy and Bayer (2019) examined nascent EEs through resource dependence theory, proposing bridging and buffering strategies for managing resource dependencies.

They suggested that ecosystem size and collaborative values influence these strategies, which in turn affect EE resilience. These studies collectively demonstrate the relevance of RBV in understanding EEs' resource dynamics, creativity, and development strategies. Assessing EEs continues to raise issues; nevertheless, recent research has established methods to quantify and evaluate ecosystem components across many areas, underscoring the systemic characteristics of entrepreneurial economies (Leendertse et al., 2021). Integrating the RBV with other theories that account for external influences is crucial for navigating changeable corporate environments. Multi-criteria decision-making methodologies have been utilized to evaluate and rank nations according to their entrepreneurial performance, highlighting the necessity of including many ecosystem factors in policy development (Popovic et al., 2024).

The EE concept has gained prominence in research and policy, particularly in the MENA region, where resource scarcity raises issues for entrepreneurs (Roomi et al., 2023). EEs are viewed as resource allocation systems that mobilize and circulate resources for new venture creation (Shi and Shi, 2022). The process of resource allocation in EEs involves four interconnected themes: resource endowments, use, dynamics, and enabling conditions (Shi and Shi, 2022). Recent studies have applied novel perspectives, such as a service-dominant logic lens, to understand value exchange within EEs (Donaldson et al., 2024). However, the EE concept faces several limitations, including a lack of clear analytical frameworks, insufficient integration of network theory, and challenges in identifying relevant institutions and spatial scales (Alvedalen and Boschmaa, 2017). Future research should address these shortcomings by adopting comparative, multi-scalar perspectives and longitudinal analysis.

Recent research on EEs highlights their complexity and the importance of understanding their structure for effective policy-making. Studies have applied various frameworks to analyze EEs, including complementarity perspectives (Belitski and Büyükbacı, 2021) and service-dominant logic (Donaldson et al., 2024). The structure of EEs can vary, with some being more centralized and others more decentralized (Brown and Mason, 2017). Key components of EEs include both actor (e.g.,

academics, businesses, and government) and non-actor elements (e.g., human capital, networks, culture) (Asmit et al., 2024). Environmental resources are particularly crucial in rural EEs, distinguishing them from urban counterparts (Asmit et al., 2024). The efficiency of EEs depends on the complementarity of services and resources (Belitski and Büyükbacı, 2021), as well as the value exchange processes among ecosystem actors (Donaldson et al., 2024).

Network analysis offers valuable insights into entrepreneurial ecosystem (EE) structures and dynamics. Studies have shown that EEs evolve over time, shifting from calculative to affective ties (Jack, 2010) and following cyclical processes of relational development (Scott et al., 2022). Longitudinal research reveals how network configurations change and impact ecosystem performance (Guéneau et al., 2023). Key network metrics, such as centrality and density, can provide snapshots of ecosystem organization, while examining tie formation and persistence helps understand ecosystem evolution (Guéneau et al., 2023). However, the EE concept still lacks a clear analytical framework and has not fully exploited network theory insights (Alvedalen and Boschmaa, 2017). For Alvedalen and Boschmaa (2017), future research should focus on developing comparative and multi-scalar perspectives, incorporating institutional factors, and systematically studying EE evolution over time. Such approaches can enhance our understanding of EE adaptability and resilience to external shocks.

Ecosystem dynamics in entrepreneurial and innovation contexts involve complex interactions among various actors, shaping resource flows and knowledge spillovers (Alaassar et al., 2022; Qian, 2018). These dynamics are influenced by factors such as collaboration, intermediation, and regulatory environments. Entrepreneurship serves as a mechanism for transmitting knowledge spillovers, contributing to regional innovation and economic development. The effectiveness of this mechanism depends on elements within the regional entrepreneurial ecosystem, including knowledge bases, networks, and culture (Qian, 2018). Social network analysis reveals the importance of connections among entrepreneurs, support organizations, and between these groups in shaping ecosystem dynamics (Motoyama and Knowlton, 2017). Understanding and orchestrating these internal dynamics is crucial for successful innovation outcomes in ecosystems, which are more fluid than traditional

networks (Paasi et al., 2023). System dynamics modeling can help identify key factors generating dynamics in innovation ecosystems.

Research indicates that collaborative networks play a crucial role in fostering entrepreneurship, particularly in the MENA region. These networks enable entrepreneurs to access resources, knowledge, and support, which are essential for innovation and business development (Van Horne et al., 2013; Zali et al., 2013). The composition and diversity of networks significantly influence venture launch time and success. However, EEs are not homogeneous and vary across regions due to factors such as economic development, cultural norms, and available support structures (Guerrero et al., 2021). In the MENA region, professional and international networks positively impact innovation, while private networks have a limited effect (Zali et al., 2013). Favorable ecosystem conditions include professional support, incubators, and R&D investments, while unfavorable conditions involve lack of funding and restrictive social norms (Guerrero et al., 2021). Understanding these ecosystem dynamics is crucial for supporting entrepreneurship at local and regional levels (Malecki, 2018).

Recent studies highlight the complex dynamics of EEs in North Africa and the Middle East. In Egypt, Morocco, and Tunisia, institutional quality plays a crucial role in sustainable firm growth, with an optimal level of institutional quality promoting substantial growth (Abozeid et al., n.d.). However, these countries face challenges such as weak regulatory frameworks and corruption. In contrast, Gulf countries like Qatar, UAE, and Saudi Arabia demonstrate more established entrepreneurship environments, with stronger institutional support (Al-Housani et al., 2023; Farid et al., 2011). Qatar, in particular, faces barriers including lack of conducive legal environments, limited access to risk-based financing, and insufficient entrepreneurial mentorship. Across Africa, critical resources such as credit access, electricity, contract enforcement, and political governance significantly impact entrepreneurship quality and ecosystem depth (Atiase et al., 2017). The studies emphasize the need for improved institutional infrastructure and balanced regulation to foster entrepreneurship development in these regions.

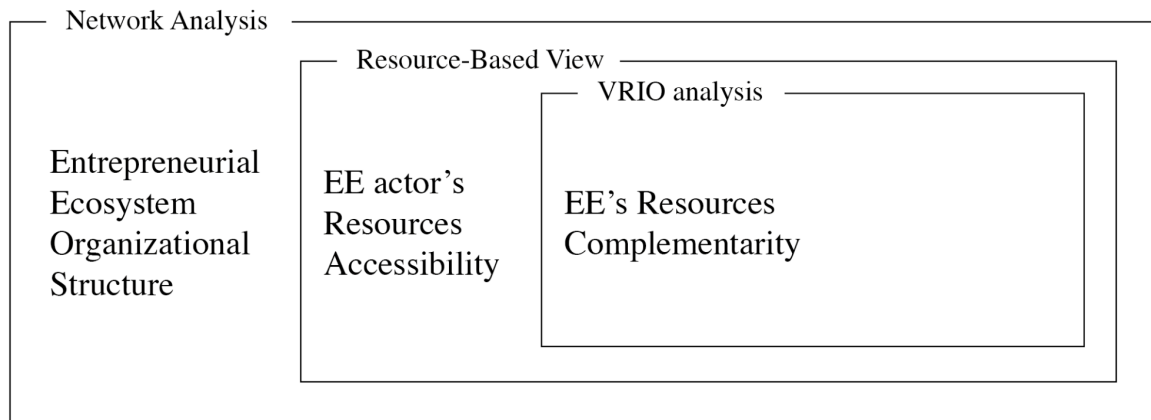
By comparing these ecosystems, researchers can identify both the strengths and weaknesses that shape entrepreneurial success across different countries in the MENA region. Understanding the regional dynamics is critical for policymakers, as it enables them to design tailored interventions that support ecosystem growth.

Recent studies have highlighted the importance of measuring EEs to assess their effectiveness and foster growth. Researchers have proposed various frameworks and metrics to evaluate EEs, including network density, centrality, and RBV indicators (Leendertse et al., 2021; Stam and van de Ven, 2019). These studies emphasize the systemic nature of EEs, with strong interdependencies among elements such as physical infrastructure, finance, formal institutions, and talent. Johnson et al. (2022) stress the importance of collaborative relationships and government roles in shaping EEs. For emerging economies, Aggarwal and Kashiramka (2025) propose a weighted framework considering regional contexts, identifying key health factors like finance availability, government programs, and infrastructure. They also highlight performance factors such as entrepreneurial perception, networking, and value creation. These comprehensive approaches enable better comparison of EEs, identification of intervention opportunities, and informed decision-making for policymakers and investors.

Methodology

This study employs a robust multi-method research design integrating network analysis, the Resource-Based View (RBV), and the VRIO framework. Network analysis is used to examine the organizational structure of entrepreneurial ecosystems by mapping interconnections among actors. The RBV is applied to assess the accessibility of critical resources available to network participants, while the VRIO framework evaluates the complementarities and modularity of these resources, providing deeper insights into competitive advantage and dynamic capabilities within the ecosystem (figure 1).

Figure 1. Research design



Before initiating data collection and analysis, it is essential to first identify and qualify the key actors within each entrepreneurial ecosystem (Leendertse et al., 2021). These actors are central to the ecosystem and consist of various entities, including entrepreneurs, investors, service providers, educational institutions, governmental bodies, and support organizations (Theodoraki and Messeghem, 2017). Entrepreneurs are individuals or teams who initiate and manage new business ventures. Investors, including venture capitalists, angel investors, government funds, and private equity firms, supply the necessary capital for startup growth. Service providers such as legal advisors, accountants, business development experts, consultants, and technology firms offer essential operational support. Educational institutions, including universities, research centers, and business schools, contribute by providing skilled talent, rigorous training, and pertinent research. Government bodies and policymakers create the regulatory and legal framework that defines the entrepreneurial environment, while support organizations like incubators, accelerators, NGOs, and entrepreneurship associations deliver various forms of assistance to entrepreneurs. We meticulously listed 1603 actors (Egypt: 314, Morocco: 210, Tunisia: 256, Qatar: 236, Saudi Arabia: 391, and UAE: 196) and qualified their activities, specialties, and services.

The authors and a research assistant use several criteria to qualify these actors. First, their relevance to the ecosystem is assessed based on their role in the entrepreneurial process, whether by providing capital, expertise, support services, or resources (Dedehayir et al., 2018). Second, we evaluate their engagement in entrepreneurial activities, ensuring they are actively involved in initiatives that

contribute directly to entrepreneurship, such as investing, mentoring, or developing programs to support new ventures. Those observations are made based on public data (website, social media activity, news). Third, based on the same observations, each actor's reputation and influence are looked at, with a focus on their involvement in major projects, network connections, and ability to change policies or practices. For this process, we chose to cross-examine the delivered data with at least a double check on every listed actor.

To identify and qualify these actors, we rely on secondary data from literature, reports, and data from local organizations like startup hubs, incubators, and chambers of commerce. We conduct direct outreach and interviews with local stakeholders to ensure a comprehensive representation of key actors when information is lacking.

Following the identification and qualification of key actors, the subsequent step involves the collection of comprehensive data on their interactions and collaborations within the entrepreneurial ecosystem. This data collection is derived from two primary sources. First, secondary data is gathered from existing reports produced by government agencies, entrepreneurship organizations, and academic studies (Von Bloh et al., 2019). These sources offer essential insights into the activities and roles of ecosystem actors, providing a foundational understanding of the ecosystem's structure. Second, we employ web scraping techniques to gather up-to-date and detailed information about the relationships among the actors. Web scraping allows the extraction of data from publicly accessible sources (Haddaway, 2015). The data gathered focuses on key elements such as collaborations and partnerships, including joint ventures, investment rounds, mentorship, and strategic alliances. We also identify the roles of various actors, including financiers, mentors, and service providers. By analyzing the co-occurrence of names in collaborative contexts, an interaction matrix is constructed to map relationships (Feicheng and Yating, 2014; Özgür et al., 2008). For the purpose of this paper, we collected 617,058 observations.

Network analysis serves as a valuable approach for investigating the structure and dynamics of EEs. It allows for the identification of relationships among various actors and the patterns of interactions that

shape the ecosystem's functionality. The process of network analysis begins with data preparation, during which a network matrix is constructed (Barabási, 2013; Barnes, 1969; Dehmer et al., 2017). This matrix serves as a representation of the ecosystem, with nodes representing actors, such as entrepreneurs, investors, and service providers, while the edges between them signify their relationships, including collaborations, investments, and partnerships. The matrix is square, with each row and column corresponding to an actor, and the matrix entries indicate the strength of the relationship between the actors.

Once the network matrix is created, it is visualized as a graph to provide a clearer understanding of the ecosystem's structure. In this graph, nodes represent actors, and edges illustrate the relationships that exist among them (Dehmer et al., 2017). By visualizing the network, key characteristics such as network density and centrality can be observed, enabling the identification of dominant actors who exert considerable influence on the ecosystem (Bloch et al., 2021; Marsden, 1993). This graph also highlights clusters of actors who are highly interconnected, as well as peripheral actors that may exhibit limited interaction with others (Lechner and Leyronas, 2007; Rattigan et al., 2007).

To further analyze the network, various metrics are calculated, offering insights into the configuration and functioning of the ecosystem. In this study, we analyze a network of actors using quantitative measures to assess structure and capacity for generating entrepreneurial opportunities (Dehmer et al., 2017; Guéneau, 2022). The “Total of Actors” comprises all nodes identified in the system, while “Active Actors” (N) are those engaged in at least one collaborative interaction. “Collaborations” (M) denote the observed ties among active participants. Density (d) is calculated as the ratio of actual collaborations to the maximum possible, indicating overall interconnection. Centrality (c) quantifies the influence of individual actors. The Cohesiveness Factor (a), defined as c multiplied by d , integrates centrality and density to capture network integration. Interconnectivity (in), computed as N divided by a , reflects the distribution of actors relative to cohesiveness, and Average Weight (aw) measures the typical intensity of interactions. The Inter-connectedness Factor (b), given by in times aw , combines these structural and weighted attributes. The Average Shortest Path (asp) represents the

mean steps between actor pairs, while the Clustering Coefficient gauges the tendency to form tightly knit groups. The Diameter Ratio (dr) normalizes the network's longest path, and its product with asp produces the Proximity Factor (c). Finally, the Entrepreneurial Opportunities Generation Factor ($a \times b \times c$) aggregates these dimensions to quantify the network's potential (Newman, 2005; Zhang and Luo, 2017).

Lastly, community detection algorithms are employed to identify clusters of actors that interact more frequently with each other than with the broader ecosystem. These communities may correspond to different sectors or subsectors within the entrepreneurial ecosystem, such as technology, finance, or healthcare (Lee and Sharma, 2024). This extensive network study elucidates the structural characteristics of the ecosystem, providing insights into its strengths, flaws, and opportunities for enhancement. This method not only improves comprehension of the ecosystem's present condition but also guides future plans for promoting collaboration and development within the entrepreneurial domain.

The RBV framework offers a solid theoretical foundation for examining the significance of resources and skills in attaining a competitive advantage within emerging economies. This study utilizes several RBV indicators to assess resource allocation, accessibility, and alignment among six MENA nations. The redundancy ratio measures resource duplication; a high ratio may suggest inefficiencies, while a low ratio might indicate a deficiency of essential resources. The accessibility index measures how easy it is for entities to get their hands on important resources like money, skills, and technology. Higher values mean easier access. The alignment ratio measures how consistent the strategic goals and values are across ecosystem players. A high alignment ratio means that there is a lot of potential for cooperation, while a low alignment ratio could reveal underlying tensions. The average pairwise complementarity measure checks how well the resources of different participants work together, and a high level of this measure means that people are working well together. The connection ratio also shows how connected all of the nodes are to each other. More connections make it easier for people to share information, resources, and opportunities.

We use the VRIO framework to assess the additional services offered within these ecosystems. The VRIO model is a conceptual analytical tool derived from the RBV that evaluates a firm's internal resources and capabilities by examining whether they are valuable, rare, difficult to imitate, and supported by an effective organizational structure (Knott, 2015). A resource meeting all these criteria is deemed capable of providing a sustainable competitive advantage, as it allows the firm to achieve superior performance in its competitive environment. This framework is widely used in strategic management research and practice to systematically identify and assess the unique assets that can drive long-term success, based on empirical evidence and theoretical developments in organizational theory and economics (Nason and Wiklund, 2018).

This approach assesses services according to four dimensions: value, scarcity, imitability, and organization. The value criteria evaluate if a service offers a competitive advantage or resolves significant entrepreneurial issues, whereas rarity measures its distinctiveness. Imitability assesses the simplicity of replication by rivals, whereas an organization analyzes the effectiveness of service structure and delivery for maximum impact.

Results

This research looks at EEs in six MENA countries: the UAE, Egypt, Tunisia, Morocco, and Saudi Arabia. It does this by using network analysis, RBV metrics, and the VRIO framework over 11 years, from 2013 to 2024. The research analyzes the structural configuration and dynamic capacities of each ecosystem, highlighting their strengths and shortcomings. It delineates essential resources and network attributes that propel entrepreneurial success, providing an in-depth comprehension of the functioning and evolution of these ecosystems.

1. Network Analysis Results

Table 1. 6 countries network structural analysis

Egypt	2013-2018	2014-2019	2015-2020	2016-2021	2017-2022	2018-2023	2019-2024
Total of Actors	314						
Active Actors - N	70	78	81	88	87	89	95
Collaboration - M	290	341	498	481	469	501	541
Density - d	0,5714	0,5714	0,5714	0,5714	0,5714	0,5714	0,5714
Centrality -c	0,1928	0,1834	0,1054	0,0964	0,0963	0,0961	0,0972
Cohesiveness factor (c*d) (a)	0,1102	0,1048	0,0603	0,0551	0,0550	0,0549	0,0555
Interconnectivity (N/a) - in	2,2581	2,5161	2,6129	2,8387	2,8065	2,8710	3,0645
Average Weight - aw	44,5207	43,6716	128,8574	131,7859	129,3881	124,7505	130,0351
Inter-connectedness factor (in*aw) (b)	100,53	109,88	336,69	374,10	363,12	358,15	398,49
Average Shortest Path (asp)	37,6542	32,0443	41,8731	45,0337	38,8407	47,2783	49,7838
Clustering Coefficient	0,3741	0,3603	0,5338	0,5325	0,5298	0,5291	0,5070
Diameter_Ratio (dr)	37,6542	32,0443	41,8731	45,0337	38,8407	47,2783	49,7838
Proximity factor (asp*dr) (c)	14,08	11,55	22,35	23,98	20,58	25,01	25,24
Entrepreneurial Opportunities generation factor (a*b*c)	156,01	132,98	453,47	494,05	411,35	491,84	558,39

Morocco	2013-2018	2014-2019	2015-2020	2016-2021	2017-2022	2018-2023	2019-2024
Total of Actors	210						
Active Actors - N	93	97	99	112	115	119	120
Collaboration - M	541	609	617	608	582	573	589
Density - d	0,4000	0,3889	0,4286	1,0000	0,3556	0,3556	0,3333
Centrality -c	0,1928	0,1834	0,1054	0,0964	0,0963	0,0961	0,0972
Cohesiveness factor (c*d) (a)	0,0771	0,0713	0,0452	0,0964	0,0343	0,0342	0,0324
Interconnectivity (N/a) - in	0,4429	0,4619	0,4714	0,5333	0,5476	0,5667	0,5714
Average Weight - aw	25,1423	27,2200	27,2269	27,7664	26,8007	26,4747	26,4194
Inter-connectedness factor (in*aw) (b)	11,13	12,57	12,84	14,81	14,68	15,00	15,10
Average Shortest Path (asp)	16,7994	12,7691	12,6553	13,8759	13,7390	12,5224	22,6756
Clustering Coefficient	0,3959	0,3821	0,3769	0,3545	0,3364	0,3358	0,3515
Diameter_Ratio (dr)	16,7994	12,7691	12,6553	13,8759	13,7390	12,5224	22,6756
Proximity factor (asp*dr) (c)	6,65	4,88	4,77	4,92	4,62	4,20	7,97
Entrepreneurial Opportunities generation factor (a*b*c)	5,71	4,38	2,77	7,02	2,32	2,15	3,90

Tunisia	2013-2018	2014-2019	2015-2020	2016-2021	2017-2022	2018-2023	2019-2024
Total of Actors	256						
Active Actors - N	86	89	90	93	93	92	88
Collaboration - M	282	297	300	306	302	296	299
Density - d	0,2444	0,3056	1,0000	1,0000	1,0000	0,2179	1,0000
Centrality -c	0,0801	0,0827	0,0455	0,0477	0,0654	0,0738	0,0913
Cohesiveness factor (c*d) (a)	0,0196	0,0253	0,0455	0,0477	0,0654	0,0161	0,0913
Interconnectivity (N/a) - in	0,3359	0,3477	0,3516	0,3633	0,3633	0,3594	0,3438
Average Weight - aw	26,1950	27,0135	26,3867	29,8889	29,6656	30,1182	34,6622
Inter-connectedness factor (in*aw) (b)	8,80	9,39	9,28	10,86	10,78	10,82	11,92
Average Shortest Path (asp)	40,9302	39,5674	35,8070	44,8621	36,2747	26,4290	28,2878
Clustering Coefficient	0,3876	0,3805	0,3644	0,3507	0,3427	0,3267	0,3632
Diameter_Ratio (dr)	40,9302	39,5674	35,8070	44,8621	36,2747	26,4290	28,2878
Proximity factor (asp*dr) (c)	15,86	15,06	13,05	15,73	12,43	8,64	10,27
Entrepreneurial Opportunities generation factor (a*b*c)	2,73	3,57	5,50	8,14	8,76	1,50	11,18

Qatar	2013-2018	2014-2019	2015-2020	2016-2021	2017-2022	2018-2023	2019-2024
Total of Actors	236						
Active Actors - N	56	55	55	51	47	40	37
Collaboration - M	150	147	134	130	114	80	64
Density - d	0,3268	0,2554	1,0000	1,0000	1,0000	0,5000	0,5000
Centrality -c	0,0905	0,1229	0,0922	0,1024	0,1059	0,1159	0,1067
Cohesiveness factor (c*d) (a)	0,0296	0,0314	0,0922	0,1024	0,1059	0,0580	0,0533
Interconnectivity (N/a) - in	0,2373	0,2331	0,2331	0,2161	0,1992	0,1695	0,1568
Average Weight - aw	8,2067	7,6939	7,9328	8,8462	8,9474	9,3625	8,2813
Inter-connectedness factor (in*aw) (b)	1,95	1,79	1,85	1,91	1,78	1,59	1,30
Average Shortest Path (asp)	7,5325	7,8236	9,6471	10,4817	8,9735	11,5854	14,5440
Clustering Coefficient	0,3963	0,3948	0,4151	0,4699	0,5007	0,4668	0,4430
Diameter_Ratio (dr)	7,5325	7,8236	9,6471	10,4817	8,9735	11,5854	14,5440
Proximity factor (asp*dr) (c)	2,99	3,09	4,00	4,93	4,49	5,41	6,44
Entrepreneurial Opportunities generation factor (a*b*c)	0,17	0,17	0,68	0,96	0,85	0,50	0,45

Saudi Arabia (KSA)	2013-2018	2014-2019	2015-2020	2016-2021	2017-2022	2018-2023	2019-2024
Total of Actors	391						
Active Actors - N	175	182	185	188	198	194	191
Collaboration - M	1151	1246	1353	1408	1463	1388	1290
Density - d	0,4000	0,8333	0,8333	1,0000	1,0000	1,0000	1,0000
Centrality -c	0,0375	0,0464	0,0327	0,0403	0,0478	0,0436	0,0463
Cohesiveness factor (c*d) (a)	0,0150	0,0387	0,0273	0,0403	0,0478	0,0436	0,0463
Interconnectivity (N/a) - in	0,4476	0,4655	0,4731	0,4808	0,5064	0,4962	0,4885
Average Weight - aw	35,0339	35,2432	39,7480	43,4339	48,7806	49,4128	50,8085
Inter-connectedness factor (in*aw) (b)	15,68	16,40	18,81	20,88	24,70	24,52	24,82
Average Shortest Path (asp)	14,0698	13,1014	12,6767	14,1511	13,8040	14,4078	18,1610
Clustering Coefficient	0,4892	0,4608	0,4853	0,4661	0,4719	0,4791	0,4903
Diameter_Ratio (dr)	14,0698	13,1014	12,6767	14,1511	13,8040	14,4078	18,1610
Proximity factor (asp*dr) (c)	6,88	6,04	6,15	6,60	6,51	6,90	8,90
Entrepreneurial Opportunities generation factor (a*b*c)	1,62	3,83	3,16	5,56	7,68	7,38	10,24

United Arab Emirates (UAE)	2013-2018	2014-2019	2015-2020	2016-2021	2017-2022	2018-2023	2019-2024
Total of Actors	196						
Active Actors - N	72	81	86	92	97	92	95
Collaboration - M	427	471	527	613	706	698	703
Density - d	0,7263	0,7095	0,9150	0,7905	0,7714	0,8129	0,7714
Centrality -c	0,0851	0,1004	0,1137	0,1051	0,0970	0,0966	0,0927
Cohesiveness factor (c*d) (a)	0,0618	0,0713	0,1041	0,0831	0,0748	0,0785	0,0715
Interconnectivity (N/a) - in	0,3673	0,4133	0,4388	0,4694	0,4949	0,4694	0,4847
Average Weight - aw	109,0867	113,3397	114,4478	120,2790	139,4688	166,9011	175,4865
Inter-connectedness factor (in*aw) (b)	40,07	46,84	50,22	56,46	69,02	78,34	85,06
Average Shortest Path (asp)	24,1084	27,3003	28,4687	27,2905	24,8692	24,0327	26,1574
Clustering Coefficient	0,4892	0,4608	0,4853	0,4661	0,4719	0,4791	0,4903
Diameter_Ratio (dr)	24,1084	27,3003	28,4687	27,2905	24,8692	24,0327	26,1574
Proximity factor (asp*dr) (c)	11,79	12,58	13,82	12,72	11,74	11,51	12,83
Entrepreneurial Opportunities generation factor (a*b*c)	29,22	41,99	72,21	59,68	60,60	70,82	78,02

The examination of EEs in several MENA nations (Table 1) uncovers unique patterns in collaboration, centrality, and opportunity creation. The ecosystem of Egypt exhibits significant growth, characterized by a consistent rise in collaboration and interconnectedness from 2013 to 2024. The Entrepreneurial Opportunities generation factor reaches a maximum of 558.39 in 2024;

nevertheless, a decrease in centrality and cohesiveness indicates that the ecosystem is becoming decentralized and fragmented over time.

The ecosystem in Morocco exhibits variable degrees of collaboration and network density. Despite a peak opportunity generation factor of 7.02 in 2022, fluctuations in cohesion and centrality suggest volatility and less sustainable growth relative to Egypt.

Tunisia has significant volatility; its opportunity generation factor peaks at 8.76 in 2022 but plummets to 1.50 in 2023, underscoring an unstable landscape characterized by rapid expansion followed by stagnation and inadequate network connectivity.

Qatar has a consistent increasing trend, with prospects reaching a peak of 10.24 in 2024 and a well-established network, but with reduced centrality. Ultimately, Saudi Arabia and the UAE possess resilient ecosystems characterized by substantial collaboration and interconnectedness, achieving notable entrepreneurial opportunities of 10.24 and 78.02 in 2024, respectively, signifying considerable potential for sustained growth.

2. RBV Metrics Results

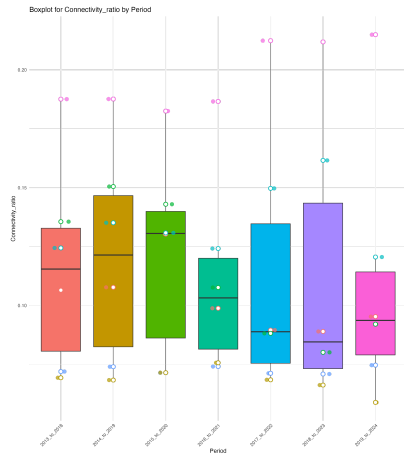
Table 2. Kruskal-Wallis Statistic on 5 structural network measures

Measure	Kruskal_Wallis_Statistic	p_value
Connectivity ratio	0.79955703211516	0.992085540081982
Accessibility index	2.44296788482836	0.874795068825538
Redundancy ratio	1.22842708164479	0.975457335150327
Avg Pairwise complementarity	1.50166112956811	0.959384140792021
Alignment ratio	0.152823920265774	0.999929775032207

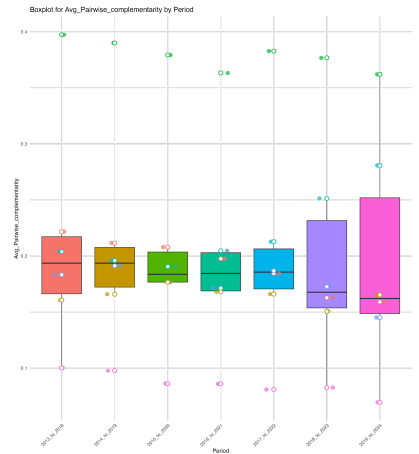
The comparative examination of the six nations' EEs reveals diverse outcomes for connection, accessibility, redundancy, complementarity, and alignment (tables 2 and 3). The connection ratio demonstrated a Kruskal-Wallis statistic of 0.7996 and a substantial p-value of 0.9921, signifying no significant disparities across the nations regarding network connectivity.

Table 3. Longitudinal boxplots comparing 6 countries RVB network measures

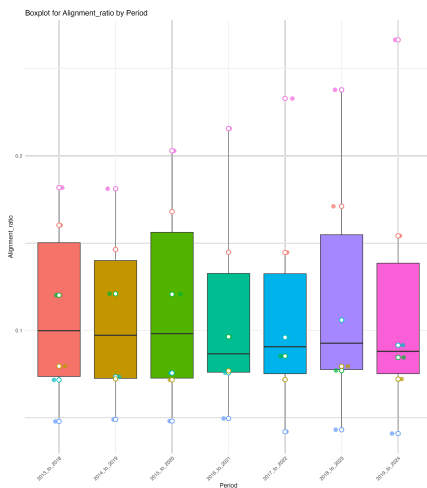
Connectivity ratio boxplot



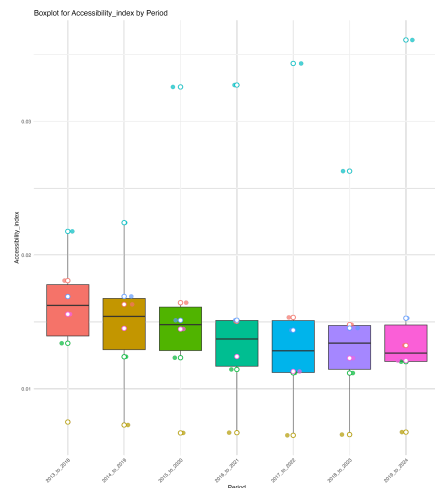
Pairwise complementarity boxplot



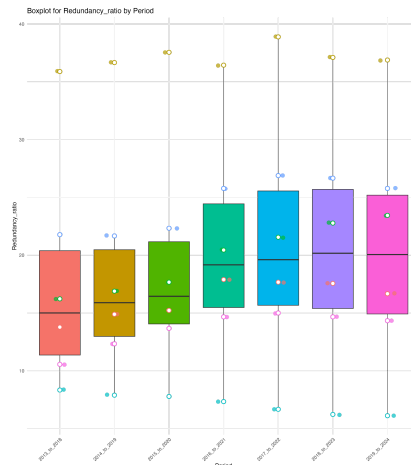
Alignment ratio boxplot



Accessibility index boxplot



Redundancy ratio boxplot



The accessibility index yielded a Kruskal-Wallis statistic of 2.4430 and a p-value of 0.8748, indicating that access to entrepreneurial resources does not substantially vary among the nations. The redundancy ratio exhibited a Kruskal-Wallis statistic of 1.2284 and a p-value of 0.9755, indicating no significant variations in redundancy among the networks. The mean pairwise complementarity yielded a Kruskal-Wallis statistic of 1.5017 and a p-value of 0.9594, signifying an absence of substantial variance in actor complementarity among nations. The alignment ratio had the maximum p-value of 0.9999, accompanied by a Kruskal-Wallis statistic of 0.1528, indicating no significant variations in the alignment of ecosystem actors among the nations.

3. VRIO Analysis of Services Complementarities

Table 4. shows the results of the Latent Class Analysis (LCA) for the different services. These results show how services differ. For example, actor.buizplanboost, actor.humanboost, and sales.protoboost, the model effectively finds two latent classes. In actor.protoboost, the model recognizes two classes, although the distribution is significantly skewed, with one class comprising 85.67% and the other just 14.33%.

Conversely, for actor.fundboost and actor.innovboost, the model did not yield significant results, presumably due to uniform values across all variables in the corresponding datasets. Likewise, actor.testboost was unable to generate appropriate classes. The AIC, BIC, and likelihood ratio statistics show that certain models, especially actor.protoboost, provide a better fit to the data compared to others, despite some challenges with specific services.

We applied the VRIO framework to assess the complementarity of services offered by ecosystem actors. The results provide valuable insights into how services within each ecosystem contribute to the overall competitive advantage.

This research provides essential insights for managers and policymakers on the structure, dynamics, and developmental prospects of growing MENA economies. Utilizing this knowledge, stakeholders may make educated decisions to promote more efficient and resilient ecosystems. The comparative

analysis highlights the significance of dense, linked networks, shown by Qatar and the UAE, in fostering entrepreneurial prospects. In contrast, the fragmented networks in Morocco, Tunisia, Saudi Arabia, and Egypt need specific actions to improve resource availability and network coherence.

Table 4. LCA on 9 Entrepreneurial Ecosystem services

LCA analysis	Service	actor.buizplanboost		actor.fundboost		actor.humanboost		actor.innovboost		actor.lawboost		actor.marketboost		actor.protoboost		sales_protoboost		actor.testboost	
	Comment	Pr(1)	Pr(2)	failed		Pr(1)	Pr(2)	all are constant		Pr(1)	Pr(2)	all are constant		Pr(1)	Pr(2)	Pr(1)	Pr(2)	failed	
IsValuable	Class 1					0,265	0,735												
	Class 2					0	1												
IsRare	Class 1					0	1			0,4501	0,5499			0,0273	0,9727	1	0		
	Class 2					0,2601	0,7399			1	0			1	0	0	1		
IsInimitable	Class 1	0,3615	0,6385							0	1			0	1	1	0		
	Class 2	1	0							0,4754	0,5246			0,9971	0,0029	0	1		
IsOrganized	Class 1	1	0													0,6316	0,3684		
	Class 2	0,1412	0,8588													1	0		
Estimated class population shares		0,2237	0,7763			0,3593	0,6407			0,6494	0,3506			0,8567	0,1433	0,9048	0,0952		
Predicted class memberships (by modal posterior prob.)		0,1429	0,8571			0,0952	0,9048			0,8333	0,1667			0,8571	0,1429	0,9048	0,0952		
Fit for 2 latent classes:																			
	number of observations:	42				42				42				42		42			
	number of estimated parameters:	5				5				5				5		7			
	residual degrees of freedom:	-2				-2				-2				-2		0			
	maximum log-likelihood:	-36,29				-31,36				-42,83				-21,79		-38,22			
	AIC(2):	82,59				72,72				95,65				53,59		90,43			
	BIC(2):	91,28				81,41				104,3				62,28		102,6			
	G ² (2) (Likelihood ratio/deviance statistic)	0				0,0000000005314				0,0000000001674				0		0			
	X ² (2) (Chi-square goodness of fit)	0				0,0000000002657				0				0		0			

Facilitating collaboration across ecosystem stakeholders, including entrepreneurs, investors, service providers, and educational institutions, is essential for improving the exchange of resources and information. The emergence of critical services like ProtoBoost, BuizPlanBoost, and SalesBoost highlights the importance of distinctive, irreplaceable, and valuable propositions, notwithstanding organizational inefficiencies. Policymakers can support these vital services by investing in innovation, improving access to financing, and offering strategic mentorship while fostering collaboration between the public and private sectors to achieve common goals.

This research provides targeted insights for developing policies that foster innovation, stimulate entrepreneurial activity, and address socio-economic challenges such as unemployment and inequality in the MENA region.

Table 5. Association rules of VRIO measures on 9 Entrepreneurial Ecosystem services

[illegible]

Conclusion

Theoretical Contribution

First, the study advances network theory by examining the dynamic structure of EEs, particularly the evolution of network centrality, connectivity, and cohesiveness over time (Huggins and Thompson, 2015; Sullivan and Ford, 2014). The study shows that strong, well-connected networks help resources flow, new ideas come up, and opportunities are created. It stresses the importance of both formal and informal networks, especially in places where resources are scarce and regulations are hard to follow.

Second, through the application of RBV metrics (Kellermanns et al., 2016; Zahra, 2021), this research deepens our understanding of the role of resources in EEs, emphasizing the distribution, accessibility, and complementarity of resources. The study observes redundancy, resource accessibility, and complementarity, which can hinder entrepreneurial success (Gruber et al., 2010). The RBV framework also sheds light on the dynamic capabilities of ecosystems, illustrating how organizations and ecosystems can reconfigure and mobilize resources in response to changing environments.

Third, the study validates its methodological approach across diverse contexts. By applying the integrated frameworks—network theory, RBV, and VRIO—to various EEs in different geographic, industrial, and regulatory settings, the research demonstrates the robustness and generalizability of its analytical models (Barile et al., 2016; Belitski and Büyükbacı, 2021). This cross-contextual validation not only confirms the reliability of the approaches used but also reveals how contextual variations shape network dynamics, resource configurations, and service complementarities (Bliemel et al., 2014).

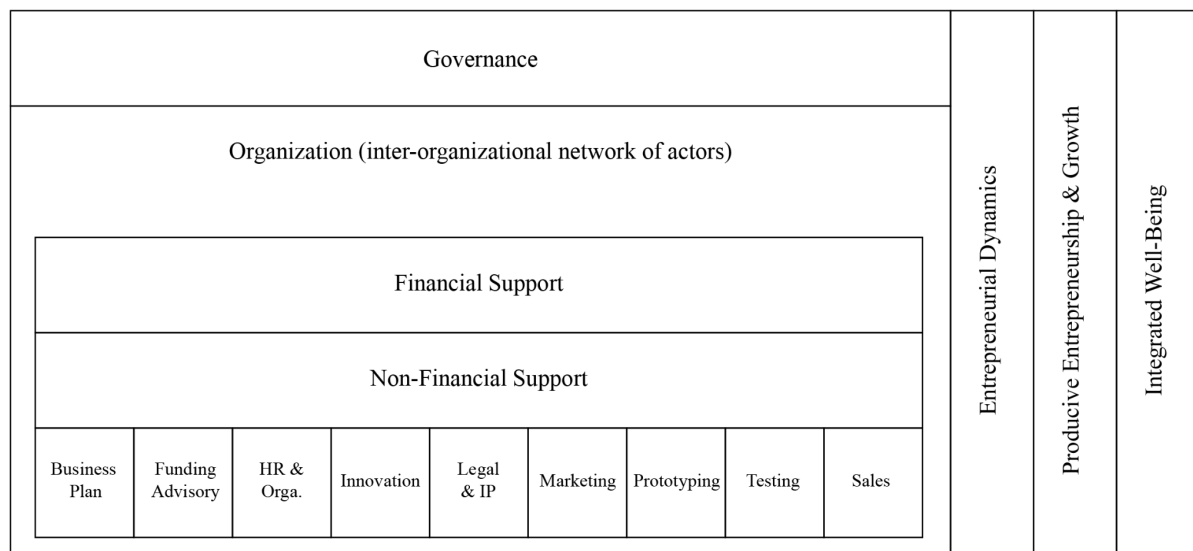
Finally, employing the VRIO framework to analyze service complementarities provides new insights into the competitive advantages offered by ecosystem services (Lingens et al., 2022). The study shows

that even unorganized services can be successful by finding services that are valuable, rare, and hard to copy. This shows how important service differentiation is for entrepreneurship.

In conclusion, this study adds to existing theories about how EEs change over time in terms of their structure, resources, and services. It also provides a complete framework for looking at how they change and compete in the MENA region.

Lastly, it enhances our comprehension of the Entrepreneurial Ecosystem Value Chain (figure 2), paving the way for future research.

Figure 2. Entrepreneurial Ecosystem Value Chain



An Entrepreneurial Ecosystem is an organizational setting composed of interconnected actors that provide both financial and non-financial assistance for entrepreneurship. This value chain integrates two primary layers. The first layer, governance, comprises formal institutions and rules that set strategic directions, establish policies, standards, and accountability, thereby coordinating activities and fostering an innovation-friendly environment. The second layer is the inter-organizational network of actors, whose diverse stakeholders offer essential supports. These supports are categorized into financial capital and non-financial “boosts” including business planning, funding advisory, HR, innovation, legal/IP, marketing, prototyping, testing, and sales support. Together, these layers

stimulate entrepreneurial dynamics, drive productive entrepreneurship, and promote long-term growth and overall ecosystem health.

Empirical Contribution

This study looks at the business environments in six MENA countries: Egypt, Tunisia, Morocco, Saudi Arabia, Qatar, and the UAE. The study offers significant insights into the structural configuration, resource dynamics, and service complementarities of each ecosystem.

The ecosystems of the UAE and Qatar are very advanced, with complex networks and easy access to resources. On the other hand, the ecosystems of Morocco and Tunisia are growing slowly, with room for improvement in how resources are distributed and how networks are connected. Despite the presence of several stakeholders in Saudi Arabia and Egypt, they have difficulties related to network fragmentation and restricted resource accessibility.

The study also points to important services—ProtoBoost, BizPlanBoost, and SalesBoost—that make it much easier for entrepreneurs to succeed because they have special, valuable qualities that can't be copied. This empirical study provides a data-driven basis for policymakers and entrepreneurs to improve ecosystem development and foster sustainable entrepreneurial growth. These findings provide direction for enhancements and future innovations.

Managerial/Policy Maker Contribution

This report provides essential insights for managers and policymakers on the structure, dynamics, and development potential of EEs in the MENA area. By utilizing these insights, stakeholders may make educated decisions to promote more efficient and resilient ecosystems. The comparative research highlights the significance of dense, linked networks, as demonstrated by Qatar and the UAE, in fostering entrepreneurial prospects. Conversely, disjointed networks in Morocco, Tunisia, Saudi Arabia, and Egypt necessitate focused interventions to bolster network coherence and augment resource accessibility.

Fostering collaboration among ecosystem participants, such as entrepreneurs, investors, service providers, and educational institutions, is crucial for enhancing the exchange of resources and information. Finding useful services like ProtoBoost, BusinessPlanBoost, and SalesBoost shows how important it is to have unique, valuable, and hard-to-copy offers, even if the organization isn't very good at them. Policymakers ought to endorse these essential services by investing in innovation, facilitating finance access, and providing strategic mentorship while enhancing collaboration between the public and private sectors towards shared objectives.

This research offers practical, area-specific insights for formulating policies that promote innovation, enhance entrepreneurial activity, and tackle socio-economic issues such as unemployment and inequality in the MENA region.

Limitations

This study offers valuable insights into EEs and the factors influencing service effectiveness, but several limitations exist. First, the analysis is based on a narrow sample of countries, which restricts the generalizability of the findings. The results may reflect characteristics unique to the selected nations and might not apply to ecosystems in regions with different economic, cultural, or institutional dynamics. Future research should expand the sample to include a wider range of countries, thereby validating the findings and assessing their global applicability.

Another limitation concerns the measurement of key variables. We used latent class analysis (LCA) to identify service attributes, but factors like IsOrganized and IsInimitable are intrinsically subjective and difficult to accurately quantify. Different industries have different ways of describing an organization's ability or inimitability. This suggests that a more refined measurement model, which could include qualitative data or expert assessments, would provide more useful information.

Finally, cross-sectional data only shows snapshots of the ecosystem and doesn't account for changes over time, making it harder to determine what caused what. Longitudinal studies are needed to understand the processual dynamics of EEs and the impact of various services over time, as well as to

explore the influence of external factors like political and economic conditions on service effectiveness.

Future Research

Future research should address current limitations and explore several key directions. Including more countries in studies will give a fuller picture of how different factors interact in different situations and help find factors that affect the effectiveness of entrepreneurial services that are specific to each region. Incorporating longitudinal data is also crucial; tracking how attributes like rarity, inimitability, value, and organization evolve over time can provide deeper insights into their impact on long-term entrepreneurial success. Longitudinal studies could further assess how external variables, such as political and regulatory shifts, shape ecosystem development.

Moreover, research should investigate the underlying mechanisms that drive service effectiveness within these ecosystems. Focusing on specific organizational practices or strategies that enhance resource inimitability and service organization can lead to more effective, tailored support for entrepreneurs. Additionally, integrating qualitative methods—such as in-depth interviews, case studies, and expert insights—will enrich our understanding of how entrepreneurs perceive and utilize these critical services. Using both quantitative and qualitative methods together will make the evidence base for future policy suggestions and service designs stronger. This will lead to more resilient and effective EEs in the long run.

In the end, this study adds to what is known about EEs in the MENA region by looking at network structures, resource allocation, and service complementarity from different angles. The findings offer both theoretical and practical insights for enhancing the effectiveness of EEs and provide a foundation for future research on the dynamics of entrepreneurship in the region.

Disclosure Statement

The authors report there are no competing interests to declare

References

- Abozeid, H.O., Elamer, A.A., Attia, E.F., n.d. Institutional Quality and Sustainable Firm Growth: Evidence From North African Countries. *Sustain. Dev.* n/a. <https://doi.org/10.1002/sd.3339>
- Aggarwal, G., Kashiramka, S., 2025. Unveiling the Dynamics of Entrepreneurial Ecosystems in an Emerging Economy: A Delphi and MCDM Approach. *Thunderbird Int. Bus. Rev.* n/a. <https://doi.org/10.1002/tie.22432>
- Alaassar, A., Mention, A.-L., Aas, T.H., 2022. Ecosystem dynamics: exploring the interplay within fintech entrepreneurial ecosystems. *Small Bus. Econ.* 58, 2157–2182. <https://doi.org/10.1007/s11187-021-00505-5>
- Al-Housani, M.I., Koç, M., Al-Sada, M.S., 2023. Investigations on Entrepreneurship Needs, Challenges, and Models for Countries in Transition to Sustainable Development from Resource-Based Economy—Qatar as a Case. *Sustainability* 15, 7537. <https://doi.org/10.3390/su15097537>
- Alvedalen, J., Boschmaa, R., 2017. A critical review of entrepreneurial ecosystems research: towards a future research agenda. *Eur. Plan. Stud.* 25, 887–903.
- Andrade, R., Pinheiro, P., Carvalho, L., Rocha, R., 2022. Building a Bridge: Knowledge Sharing Flows into Entrepreneurial Ecosystems. *J. Open Innov. Technol. Mark. Complex.* 8, 144. <https://doi.org/10.3390/joitmc8030144>
- Asmit, B., Simatupang, T.M., Rudito, B., Novani, S., 2024. Uncovering the building blocks of rural entrepreneurship: A comprehensive framework for mapping the components of rural entrepreneurial ecosystems. *Heliyon* 10. <https://doi.org/10.1016/j.heliyon.2024.e24139>
- Atiase, V.Y., Mahmood, S., Wang, Y., Botchie, D., 2017. Developing entrepreneurship in Africa: investigating critical resource challenges. *J. Small Bus. Enterp. Dev.* 25, 644–666. <https://doi.org/10.1108/JSBED-03-2017-0084>
- Audretsch, D., Mason, C., Miles, M.P., O'Connor, A., 2021. Time and the dynamics of entrepreneurial ecosystems. *Entrep. Reg. Dev.* 33, 1–14. <https://doi.org/10.1080/08985626.2020.1734257>
- Azzone, G., Bertelè, U., Rangone, A., 1995. Measuring resources for supporting resource-based competition. *Manag. Decis.* 33, 57–62. <https://doi.org/10.1108/00251749510099008>
- Balcilar, M., Elsayed, A.H., Hammoudeh, S., 2023. Financial connectedness and risk transmission among MENA countries: Evidence from connectedness network and clustering analysis1. *J. Int. Financ. Mark. Inst. Money* 82, 101656. <https://doi.org/10.1016/j.intfin.2022.101656>
- Barabási, A.-L., 2013. Network science. *Philos. Trans. R. Soc. Math. Phys. Eng. Sci.* 371, 20120375. <https://doi.org/10.1098/rsta.2012.0375>
- Barile, S., Lusch, R., Reynoso, J., Saviano, M., Spohrer, J., 2016. Systems, networks, and ecosystems in service research. *J. Serv. Manag.* 27, 652–674. <https://doi.org/10.1108/JOSM-09-2015-0268>
- Barnes, J.A., 1969. Graph Theory and Social Networks: A Technical Comment on Connectedness and Connectivity. *Sociology* 3, 215–232. <https://doi.org/10.1177/003803856900300205>
- Belitski, M., Büyükbacı, P., 2021. Uncharted waters of the entrepreneurial ecosystems research: Comparing Greater Istanbul and Reading ecosystems. *Growth Change* 52, 727–750. <https://doi.org/10.1111/grow.12457>
- Bliemel, M.J., McCarthy, I.P., Maine, E.M.A., 2014. An Integrated Approach to Studying Multiplexity in Entrepreneurial Networks. *Entrep. Res. J.* 4, 367–402. <https://doi.org/10.1515/erj-2014-0007>
- Bloch, F., Jackson, M.O., Tebaldi, P., 2021. Centrality Measures in Networks (No. arXiv:1608.05845). *arXiv*. <https://doi.org/10.48550/arXiv.1608.05845>
- Brown, R., Mason, C., 2017. Looking inside the spiky bits: a critical review and conceptualisation of entrepreneurial ecosystems. *Small Bus. Econ.* 49, 11–30. <https://doi.org/10.1007/s11187-017-9865-7>
- Burt, R.S., Soda, G., 2021. Network Capabilities: Brokerage as a Bridge Between Network Theory and the Resource-Based View of the Firm. *J. Manag.* 47, 1698–1719. <https://doi.org/10.1177/0149206320988764>

- Chen, M.-J., Michel, J.G., Lin, W., 2021. Worlds Apart? Connecting Competitive Dynamics and the Resource-Based View of the Firm. *J. Manag.* 47, 1820–1840.
<https://doi.org/10.1177/01492063211000422>
- Content, J., Bosma, N., Jordaan, J., Sanders, M., 2020. Entrepreneurial ecosystems, entrepreneurial activity and economic growth: new evidence from European regions. *Reg. Stud.*
- Credit, K., Mack, E.A., Mayer, H., 2018. State of the field: Data and metrics for geographic analyses of entrepreneurial ecosystems. *Geogr. Compass* 12, e12380.
<https://doi.org/10.1111/gec3.12380>
- Cuthbertson, R.W., Furseth, P.I., 2022. Digital services and competitive advantage: Strengthening the links between RBV, KBV, and innovation. *J. Bus. Res.* 152, 168–176.
<https://doi.org/10.1016/j.jbusres.2022.07.030>
- Dedehayir, O., Mäkinen, S.J., Roland Ortt, J., 2018. Roles during innovation ecosystem genesis: A literature review. *Technol. Forecast. Soc. Change* 136, 18–29.
<https://doi.org/10.1016/j.techfore.2016.11.028>
- Dehmer, M., Emmert-Streib, F., Shi, Y., 2017. Quantitative Graph Theory: A new branch of graph theory and network science. *Inf. Sci.* 418–419, 575–580.
<https://doi.org/10.1016/j.ins.2017.08.009>
- Donaldson, C., Newbery, R., Mirabent, J.B., Kallmuenzer, A., 2024. Decoding value exchange in entrepreneurial ecosystems through a service-dominant lens. *Small Bus. Econ.*
<https://doi.org/10.1007/s11187-024-00935-x>
- Farid, M.I., ElSayed-Elkhouly, S.M., Barnes, M.J., 2011. Institutional context for entrepreneurship in Arab countries. *Int. J. Sustain. Soc.* 3, 292–311. <https://doi.org/10.1504/IJSSOC.2011.041269>
- Feicheng, M., Yating, L., 2014. Utilising social network analysis to study the characteristics and functions of the co-occurrence network of online tags. *Online Inf. Rev.* 38, 232–247.
<https://doi.org/10.1108/OIR-11-2012-0124>
- Godley, A., Morawetz, N., Soga, L., 2021. The complementarity perspective to the entrepreneurial ecosystem taxonomy. *Small Bus. Econ.* 56, 723–738.
<https://doi.org/10.1007/s11187-019-00197-y>
- Granovetter, M., 1985. Economic Action and Social Structure: The Problem of Embeddedness. *Am. J. Sociol.* 91. <https://doi.org/10.1086/228311>
- Gruber, M., Heinemann, F., Brettel, M., Hungeling, S., 2010. Configurations of resources and capabilities and their performance implications: an exploratory study on technology ventures. *Strateg. Manag. J.* 31, 1337–1356. <https://doi.org/10.1002/smj.865>
- Guéneau, G., 2022. Entrepreneurial ecosystems structure, a measurement framework and first findings in african low income countries (These de doctorat). Paris 1.
- Guéneau, G., Chabaud, D., Chalus Sauvannet, M.-C., 2023. Sticky ties: Quest for structural inter-organizational configurations in entrepreneurial ecosystems. *Technol. Forecast. Soc. Change* 190, 122416. <https://doi.org/10.1016/j.techfore.2023.122416>
- Gueneau, G., Chabaud, D., Chalus-sauvannet, M.-C., 2021. Entrepreneurial Ecosystems Power of Networks : central role impact in low income countries. *Acad. Manag. Proc.* 2021, 13891.
<https://doi.org/10.5465/AMBPP.2021.13891abstract>
- Guerrero, M., Liñán, F., Cáceres-Carrasco, F.R., 2021. The influence of ecosystems on the entrepreneurship process: a comparison across developed and developing economies. *Small Bus. Econ.* 57, 1733–1759. <https://doi.org/10.1007/s11187-020-00392-2>
- Haddaway, N.R., 2015. The Use of Web-scraping Software in Searching for Grey Literature. *Grey J.* 11, 5.
- Huggins, R., Thompson, P., 2015. Entrepreneurship, innovation and regional growth: a network theory. *Small Bus. Econ.* 45, 103–128. <https://doi.org/10.1007/s11187-015-9643-3>
- Jack, S.L., 2010. Approaches to studying networks: Implications and outcomes. *J. Bus. Ventur.* 25, 120–137. <https://doi.org/10.1016/j.jbusvent.2008.10.010>
- Johnson, E., Hemmatian, I., Lanahan, L., Joshi, A.M., 2022. A Framework and Databases for Measuring Entrepreneurial Ecosystems. *Res. Policy* 51, 104398.
<https://doi.org/10.1016/j.respol.2021.104398>
- Kellermanns, F., Walter, J., Crook, T.R., Kemmerer, B., Narayanan, V., 2016. The Resource-Based

- View in Entrepreneurship: A Content-Analytical Comparison of Researchers' and Entrepreneurs' Views: The Resource-Based View in Entrepreneurship: A Content-Analytical Comparison of Researchers' and Entrepreneurs' Views. *J. Small Bus. Manag.* 54, 26–48. <https://doi.org/10.1111/jsbm.12126>
- Knott, P.J., 2015. Does VRIO help managers evaluate a firm's resources? *Manag. Decis.* 53, 1806–1822. <https://doi.org/10.1108/MD-08-2014-0525>
- Komlósi, É., Sebestyén, T., Tóth-Pajor, Á., Bedő, Z., 2022. Do specific entrepreneurial ecosystems favor high-level networking while others not? Lessons from the Hungarian IT sector. *Technol. Forecast. Soc. Change* 175, 121349. <https://doi.org/10.1016/j.techfore.2021.121349>
- Korsgaard, S., Wigren-Kristoferson, C., Brundin, E., Hellerstedt, K., Alsos, G.A., Grande, J., 2022. Entrepreneurship and embeddedness: process, context and theoretical foundations. *Entrep. Reg. Dev.* 34, 210–221. <https://doi.org/10.1080/08985626.2022.2055152>
- Lechner, C., Leyronas, C., 2007. Network-centrality versus network-position in regional networks: what matters most? a study of a French high-tech cluster. *Int. J. Technoentrepreneurship* 1, 78. <https://doi.org/10.1504/IJTE.2007.013271>
- Lee, M., Sharma, P., 2024. Leveraging the Louvain algorithm for enhanced group formation and collaboration in online learning environments. *Int. J. Educ. Technol. High. Educ.* 21, 65. <https://doi.org/10.1186/s41239-024-00495-w>
- Leendertse, J., Schrijvers, M., Stam, E., 2021. Measure Twice, Cut Once: Entrepreneurial Ecosystem Metrics. *Res. Policy* 104336. <https://doi.org/10.1016/j.respol.2021.104336>
- Lingens, B., Seeholzer, V., Gassmann, O., 2022. The architecture of innovation: how firms configure different types of complementarities in emerging ecosystems. *Ind. Innov.* 29, 1108–1139. <https://doi.org/10.1080/13662716.2022.2123307>
- Mack, E., Mayer, H., 2016. The evolutionary dynamics of entrepreneurial ecosystems. *Urban Stud.* 53, 2118–2133. <https://doi.org/10.1177/0042098015586547>
- Mago, S., van der Merwe, S., 2023. Exploring Entrepreneurial Ecosystems in Developed Countries: A Systematic Review. *Sage Open* 13, 21582440231217886. <https://doi.org/10.1177/21582440231217886>
- Malecki, E.J., 2018. Entrepreneurship and entrepreneurial ecosystems. *Geogr. Compass* 12, e12359. <https://doi.org/10.1111/gec3.12359>
- Marsden, P.V., 1993. The reliability of network density and composition measures. *Soc. Netw.* 15, 399–421. [https://doi.org/10.1016/0378-8733\(93\)90014-C](https://doi.org/10.1016/0378-8733(93)90014-C)
- McGee, J., 2015. Resource-Based View, in: Cooper, C.L. (Ed.), *Wiley Encyclopedia of Management*. Wiley, pp. 1–8. <https://doi.org/10.1002/9781118785317.weom120134>
- Meshram, S.A., Rawani, A.M., 2019. Understanding Entrepreneurial Ecosystem. *Int. J. Soc. Ecol. Sustain. Dev. IJSESD* 10, 103–115. <https://doi.org/10.4018/IJSESD.2019070107>
- Motoyama, Y., Knowlton, K., 2017. Examining the Connections within the Startup Ecosystem: A Case Study of St. Louis. *Entrep. Res. J.* 7. <https://doi.org/10.1515/erj-2016-0011>
- Nason, R.S., Wiklund, J., 2018. An Assessment of Resource-Based Theorizing on Firm Growth and Suggestions for the Future. *J. Manag.* 44, 32–60. <https://doi.org/10.1177/0149206315610635>
- Newbert, S.L., 2014. Assessing performance measurement in RBV research. *J. Strategy Manag.* 7, 265–283. <https://doi.org/10.1108/JSMA-10-2013-0061>
- Newman, M.E.J., 2005. A measure of betweenness centrality based on random walks. *Soc. Netw.* 27, 39–54. <https://doi.org/10.1016/j.socnet.2004.11.009>
- Özgür, A., Cetin, B., Bingol, H., 2008. Co-occurrence network of Reuters news. *Int. J. Mod. Phys. C* 19, 689–702. <https://doi.org/10.1142/S0129183108012431>
- Paasi, J., Wiman, H., Apilo, T., Valkokari, K., 2023. Modeling the dynamics of innovation ecosystems. *Int. J. Innov. Stud.* 7, 142–158. <https://doi.org/10.1016/j.ijis.2022.12.002>
- Popovic, G., Fedajev, A., Mitic, P., Meidute-Kavaliauskiene, I., 2024. An ADAM-based approach to unveiling entrepreneurial ecosystems in selected European countries. *Manag. Decis.* ahead-of-print. <https://doi.org/10.1108/MD-12-2023-2420>
- Powell, W.W., Koput, K.W., Smith-Doerr, L., 1996. Interorganizational Collaboration and the Locus of Innovation: Networks of Learning in Biotechnology. *Adm. Sci. Q.* 41, 116. <https://doi.org/10.2307/2393988>

- Prokop, D., Thompson, P., 2022. Defining networks in entrepreneurial ecosystems: the openness of ecosystems. *Small Bus. Econ.* <https://doi.org/10.1007/s11187-022-00710-w>
- Qian, H., 2018. Knowledge-Based Regional Economic Development: A Synthetic Review of Knowledge Spillovers, Entrepreneurship, and Entrepreneurial Ecosystems. *Econ. Dev. Q.* 32, 163–176. <https://doi.org/10.1177/0891242418760981>
- Queissner, M., Stolz, L., Weiss, M., 2024. A meta-analysis of entrepreneurial ecosystem elements and entrepreneurial activity. *Small Bus. Econ.* <https://doi.org/10.1007/s11187-024-00953-9>
- Rattigan, M.J., Maier, M., Jensen, D., 2007. Graph clustering with network structure indices, in: *Proceedings of the 24th International Conference on Machine Learning - ICML '07*. Presented at the the 24th international conference, ACM Press, Corvallis, Oregon, pp. 783–790. <https://doi.org/10.1145/1273496.1273595>
- Roomi, M.A., Coduras, A., Saiz-Alvarez, J.M., 2023. Riyadh: Evolving to Become as One of the MENA Region's Key Entrepreneurial Ecosystems. *Sustainability* 15, 15109. <https://doi.org/10.3390/su152015109>
- Roundy, P.T., Bayer, M.A., 2019. To bridge or buffer? A resource dependence theory of nascent entrepreneurial ecosystems. *J. Entrep. Emerg. Econ.* 11, 550–575. <https://doi.org/10.1108/JEEE-06-2018-0064>
- Scott, S., Hughes, M., Ribeiro-Soriano, D., 2022. Towards a network-based view of effective entrepreneurial ecosystems. *Rev. Manag. Sci.* 16, 157–187. <https://doi.org/10.1007/s11846-021-00440-5>
- Shi, X., Shi, Y., 2022. Unpacking the process of resource allocation within an entrepreneurial ecosystem. *Res. Policy, Uncommon Methods and Metrics for Local Entrepreneurial Ecosystems* 51, 104378. <https://doi.org/10.1016/j.respol.2021.104378>
- Spigel, B., Harrison, R., 2018. Toward a process theory of entrepreneurial ecosystems. *Strateg. Entrep. J.* 12, 151–168. <https://doi.org/10.1002/sej.1268>
- Stam, E., 2015. Entrepreneurial Ecosystems and Regional Policy: A Sympathetic Critique. *Eur. Plan. Stud.* 23, 1759–1769. <https://doi.org/10.1080/09654313.2015.1061484>
- Stam, E., van de Ven, A., 2019. Entrepreneurial ecosystem elements. *Small Bus. Econ.* 56, 809–832. <https://doi.org/10.1007/s11187-019-00270-6>
- Sullivan, D.M., Ford, C.M., 2014. How Entrepreneurs use Networks to Address Changing Resource Requirements during Early Venture Development. *Entrep. Theory Pract.* 38, 551–574. <https://doi.org/10.1111/etap.12009>
- Szerb, L., Lafuente, E., Horváth, K., Páger, B., 2019. The relevance of quantity and quality entrepreneurship for regional performance: the moderating role of the entrepreneurial ecosystem. *Reg. Stud.* 53, 1308–1320. <https://doi.org/10.1080/00343404.2018.1510481>
- Theodoraki, C., Messegem, K., 2017. Exploring the entrepreneurial ecosystem in the field of entrepreneurial support: a multi-level approach. *Int. J. Entrep. Small Bus.* 31, 47–66. <https://doi.org/10.1504/IJESB.2017.083847>
- Thompson-Whiteside, H., Fletcher-Brown, J., Middleton, K., Turnbull, S., 2023. Emergence in emergency: How actors adapt to service ecosystem disruption. *J. Bus. Res.* 162, 113800. <https://doi.org/10.1016/j.jbusres.2023.113800>
- Van Horne, C., Belkacem, L., Al Fusail, T.A., 2013. Composition of networks around entrepreneurs: cross-national comparisons in the Middle East and North Africa. *Int. J. Bus. Glob.* 11, 337–352. <https://doi.org/10.1504/IJBG.2013.056872>
- Vedula, S., Kim, P.H., 2019. Gimme shelter or fade away: the impact of regional entrepreneurial ecosystem quality on venture survival. *Ind. Corp. Change* 28, 827–854. <https://doi.org/10.1093/icc/dtz032>
- Von Bloh, J., Broekel, T., Özgün, B., Sternberg, R., 2019. New(s) data for entrepreneurship research? An innovative approach to use Big Data on media coverage. *Small Bus. Econ.* <https://doi.org/10.1007/s11187-019-00209-x>
- Wigren-Kristoferson, C., Brundin, E., Hellerstedt, K., Stevenson, A., Aggestam, M., 2022. Rethinking embeddedness: a review and research agenda. *Entrep. Reg. Dev.* 1–25. <https://doi.org/10.1080/08985626.2021.2021298>
- Zahra, S.A., 2021. The Resource-Based View, Resourcefulness, and Resource Management in Startup

- Firms: A Proposed Research Agenda. *J. Manag.* 47, 1841–1860.
<https://doi.org/10.1177/01492063211018505>
- Zali, M.R., Bastian, B.L., Qureshi, S., 2013. Promoting innovation in the MENA region: the role of social norms and individual factors in entrepreneurial networks. *Int. J. Bus. Glob.* 11, 413–426. <https://doi.org/10.1504/IJBG.2013.056877>
- Zengyu Huang, V., Nandialath, A., Kassim Alsayaghi, A., Esra Karadeniz, E., 2013. Socio-demographic factors and network configuration among MENA entrepreneurs. *Int. J. Emerg. Mark.* 8, 258–281. <https://doi.org/10.1108/17468801311330329>
- Zhang, J., Luo, Y., 2017. Degree Centrality, Betweenness Centrality, and Closeness Centrality in Social Network, in: *Proceedings of the 2017 2nd International Conference on Modelling, Simulation and Applied Mathematics (MSAM2017)*. Presented at the 2017 2nd International Conference on Modelling, Simulation and Applied Mathematics (MSAM2017), Atlantis Press, Bankok, Thailand. <https://doi.org/10.2991/msam-17.2017.68>